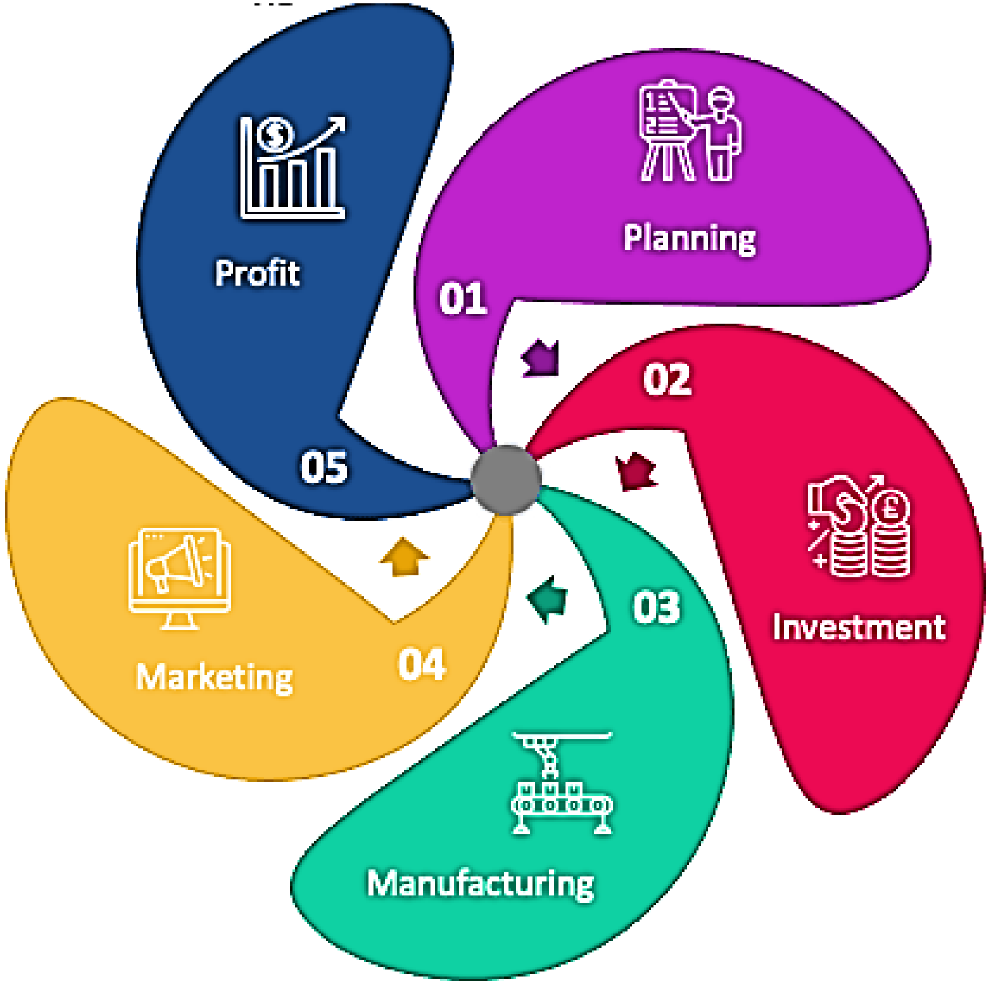

بسم الله الرحمن الرحيم
In the name of Allah, Most Gracious,
Most Merciful.
MANSOOR

Engineering Economics (HS - 407)



Course Instructor

Engr. Yasir Hamid
Lecturer
Department of Mechanical Engg.
HITEC University, Taxila Cantt

Contact Details

+92-334-5553363
yasir.hamid@hitecuni.edu.pk
yasir.hamid2k09@gmail.com

Course Outline & Reference Books

Join Google Classroom:

https://classroom.google.com/c/NzYwNDU5NjkwNDg2?cjc=rk4peiq

rk4peiq

Course Outline & Reference Books

Reference Books

1. Engineering Economy by William G. Sullivan, Elin W. Wicks, C. Patrick Koelling

2. Economics by Campbell R Cornell, Stanley L. Bruce

3. Project Management, the managerial process by Erik W. Larson, Clifford F. Gray

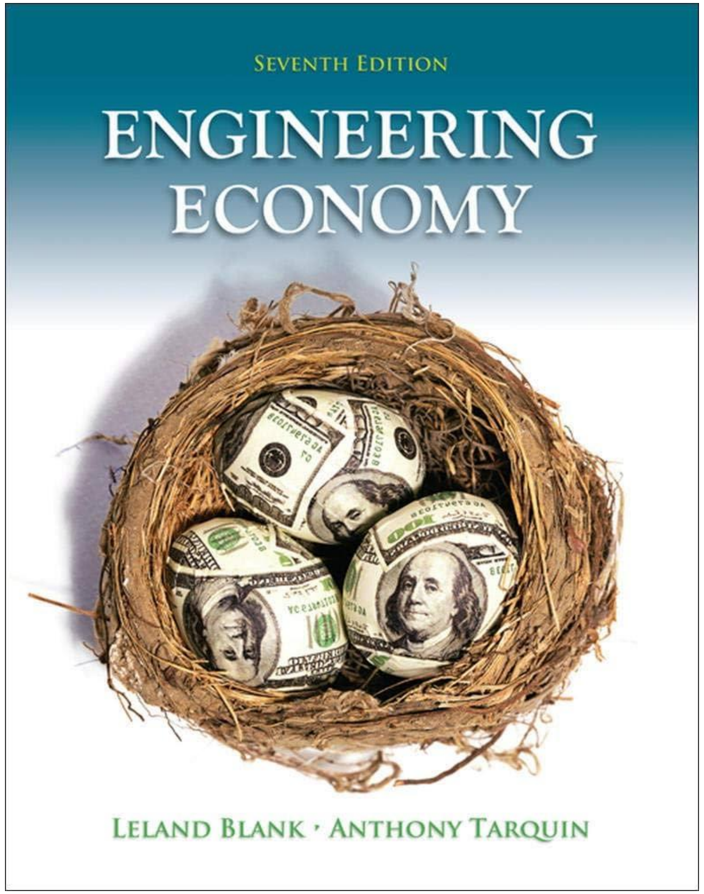
4. Karl E. Case's Principles of Economics (10th Edition)

Office Visiting Hours

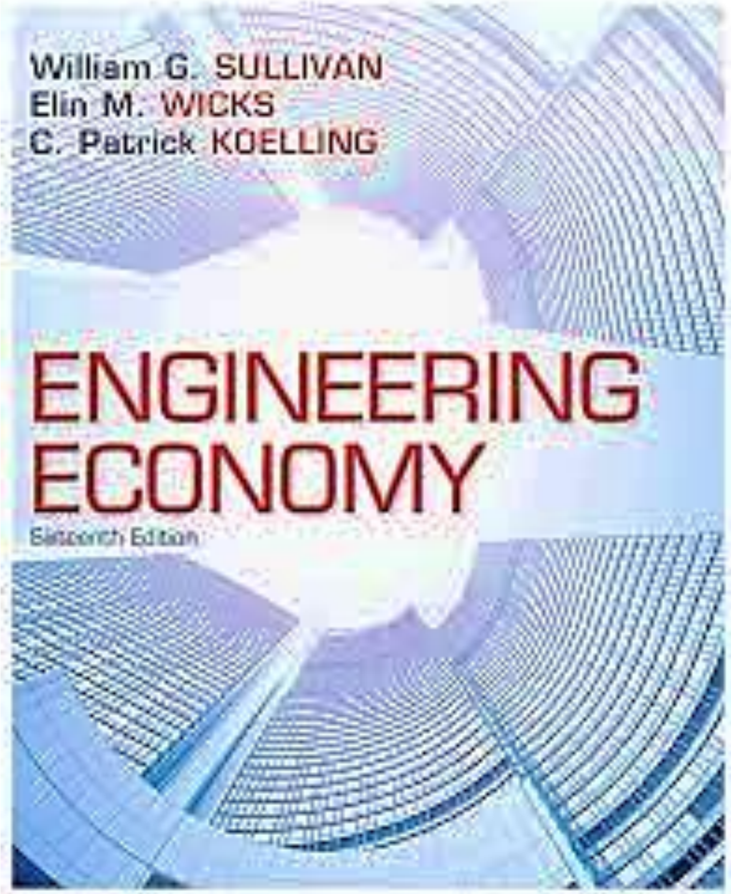
- Office 08 MED,
- Phone: Ext 327/329
- Office Hours: 0830 - 1530 hrs.
- Lunch Break 1230 - 1330 hrs.
- Student Hours: 0930 - 1630 hrs. Tuesday

Course Outline & Reference Books

Reference Book 1



Reference Book 2



Reference Book 3

Course Outline & Grading Policy

Assignments=10% Quizzes=05% for 1/CH → 2 Quizzes, Project=10% Mid Exam=30% each, Finals=45%

Quiz 1 . . . . .Week 03
Assignment 1 . . . . .Week 04
Quiz 2. . . . . .Week 05
Assignment 2 . . . . .Week 06

Mid Exam . . . . Week 08
Quiz 3 . . . . .Week 10
Assignment 3 . . . . .Week 11
Quiz 4 . . . . .Week 12
Assignment 4 . . . . .Week 13
Project . . . . .Week 14
Presentation. . . . . .Week 15

Finals . . . . Week 17

Course Outline & Class Policy

- You are highly encouraged to ask questions and discuss the course material with me and your classmates.
- No cheating, fabrication, falsification, forgery and computer misuse.
- Please switch off/put on silent your cell phones.
- Adhere to the given date for assignment submission.
- Academic Honesty: Plagiarism will not be tolerated at any level.
- Extra Help: Do not hesitate to come to my office during office hours or by appointment to discuss a problem or any aspect of the course.
- University Attendance Policy: Students are expected to attend classes regularly. Minimum attendance must be 75%.

Course Outline & Important Note

Make-Up Of Any Exams / Quizzes / Sessional Will Not Be Entertained – Make Sure To Attend It As Scheduled

Make Sure To Comply With The Attendance Policy (75% minimum). No Relaxation Will Be Granted

Make Habit of Self-Study – Make Habit of Reading Books
Make Use of the Power of Internet for Research

Course HS - 402, Engineering Economics, CR 1, (2+0)/week

Course Outline & Important Note

Module	Topic	References	Week/Lecture
I	**Introduction - Basics of Economics** Economics and Market Forces Coordinates of Economics Scarcity, Choice, Opportunity Cost Marginal Benefit, Marginal Cost	Textbook 2 (Chap. 1)	1
II	**Interest Rate, Cost of Money** Time Value of Money Methods of Depreciation Cost Benefit analysis, Concepts of PW, FW, IRR, etc	Related portions Textbook 1 (Chap 4 & Chap 5), Ref Book 1 (Chapter 1)	2, 3

Course Outline & Important Note

III	**Market, Demand, Supply, Circular Flow Diagram and Theory of Firms Behaviour** Types of Markets Scarcity, Choice, Opportunity Cost Demand Law Supply Law Effect of Determinants in Demand Supply Curve Shift Price Equilibrium Stakeholders Theory of Firms Behaviour Production Functions	Textbook 4 (Chap. 2 & 3)	4,5
IV	**Cost Terminologies and Breakeven Analysis** Cost Terminologies Cost Curves Breakeven Analysis	Textbook 1 (Chap. 2)	6

Course Outline & Important Note

V	**Decision Making Tools** SWOT Analysis Six Thinking Hats Force Field Analysis Ishikawa Diagram Decision Matrix	Textbook 1 (Chap. 1)	7,8
VI	**Project Management** Planning & Building up Project Types of Projects Work Breakdown Structure Activity Network Budgeting a project through costing components Earned Value Analysis	Related portions Textbook 3 (Chap 4, 5 & 6)	9,10
VII	**Pricing Strategies** Pricing Policy Pricing Objective		11

Course Outline & Important Note

VIII	**<u>Macro-Economics</u>** Introduction to Macro-Economics Public Finance: The Economics of Taxation Market Imperfections and the Role of Government, Monopoly and Antitrust Laws Income Distribution and Poverty	Textbook 4 (Chap 13, 18, 19, 20, 22)	12, 13, 14 15
IX	Project + Presentation		16

Course HS - 402, Engineering Economics, CR 1, (2+0)/week

Course Outline & Course Learning Outcomes (CLO)

CLO 1: Compute cost analysis, demand, supply, breakeven analysis of an engineering work and develop skills for cost benefit analysis of projects and running operations. (Cognitive – C3 – Apply)

CLO 2: Analyze different alternatives and generate a solution keeping in view of the profit maximization, cost reduction concepts for ongoing operations and projects (Cognitive – C6 – Generate)

CLO 3: Explain the different economic principles applied in businesses, manufacturing, and service concerns. (Cognitive – C2 – Understand)

The course is designed so that students will achieve the PLO/s:

PLO-2: Problem Analysis: An ability to identify, formulate research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

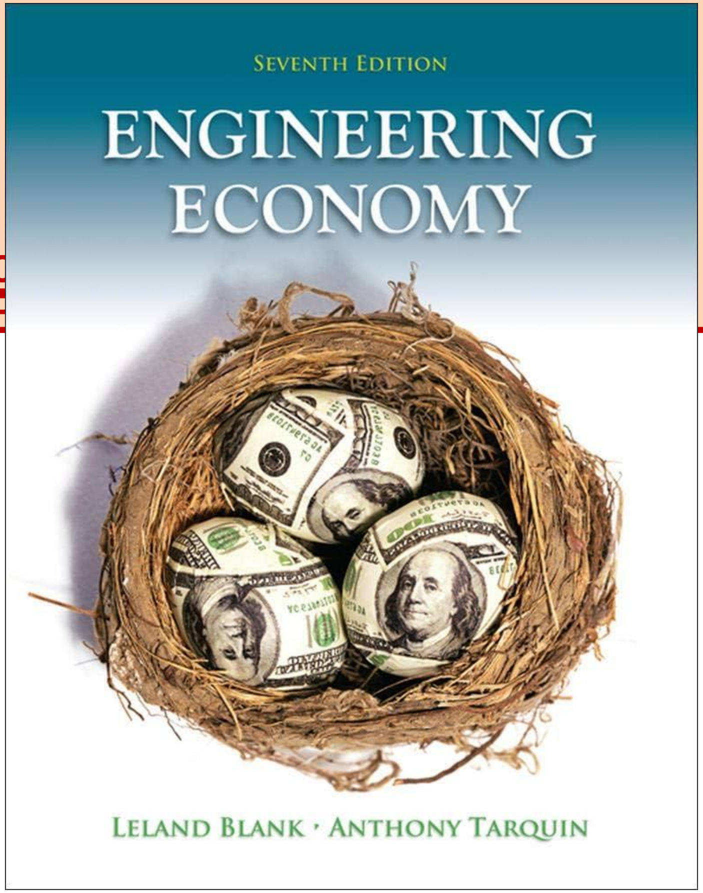
PLO-11: Project Management: An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and or leader in a team, to manage projects in a multidisciplinary environment.

PLO-12: Life-long Learning: An ability to recognize the need for, and have the preparation and ability to engage in independent and lifelong learning.

Introduction to Engineering Economics

Chapter # (1)

The Scope and Nature of Engineering



The Foundation of Engineering Economics by Leland Blank

Introduction to Engineering Economics

The Study of the Choices People Make to Satisfy Their Wants and Needs with a Limited Supply of Resources

Introduction to Engineering Economics

Economics is a social science directed at the satisfaction of **needs** and **wants** through the allocation of scarce resources which have alternative uses.

Introduction to Engineering Economics

Economics

[ˌe-kə-ˈnä-miks]

A social science concerned chiefly with description and analysis of the production, distribution, and consumption of goods and services.

Introduction to Engineering Economics

Economics

PRODUCTION

DISTRIBUTION

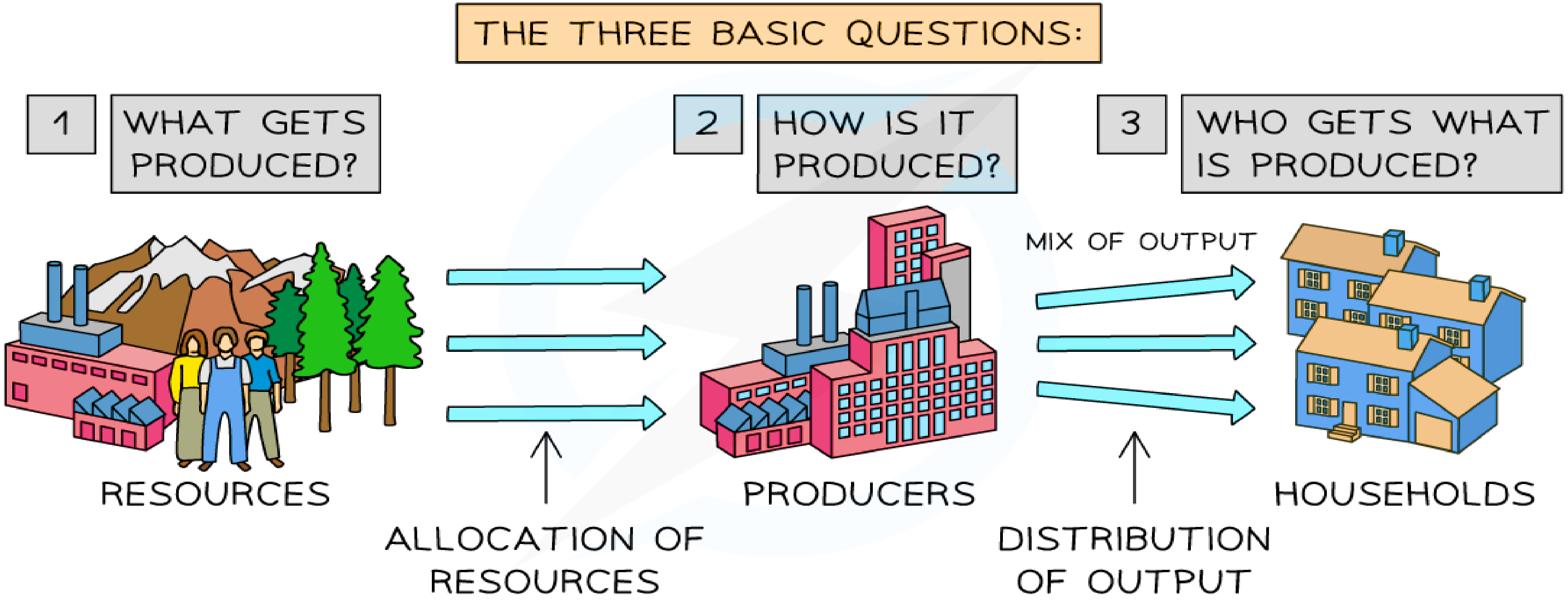
CONSUMPTION

Goods

Services

The Economic Problem: Scarcity & Choice

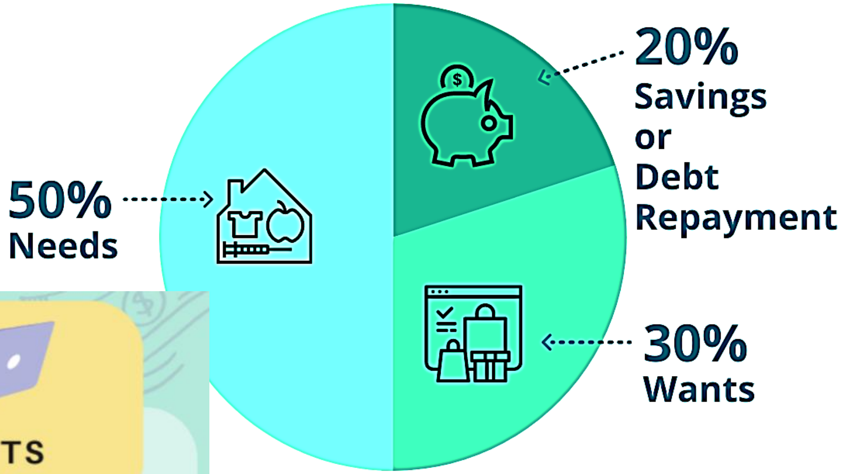
PRODUCTION **DISTRIBUTION** **CONSUMPTION**

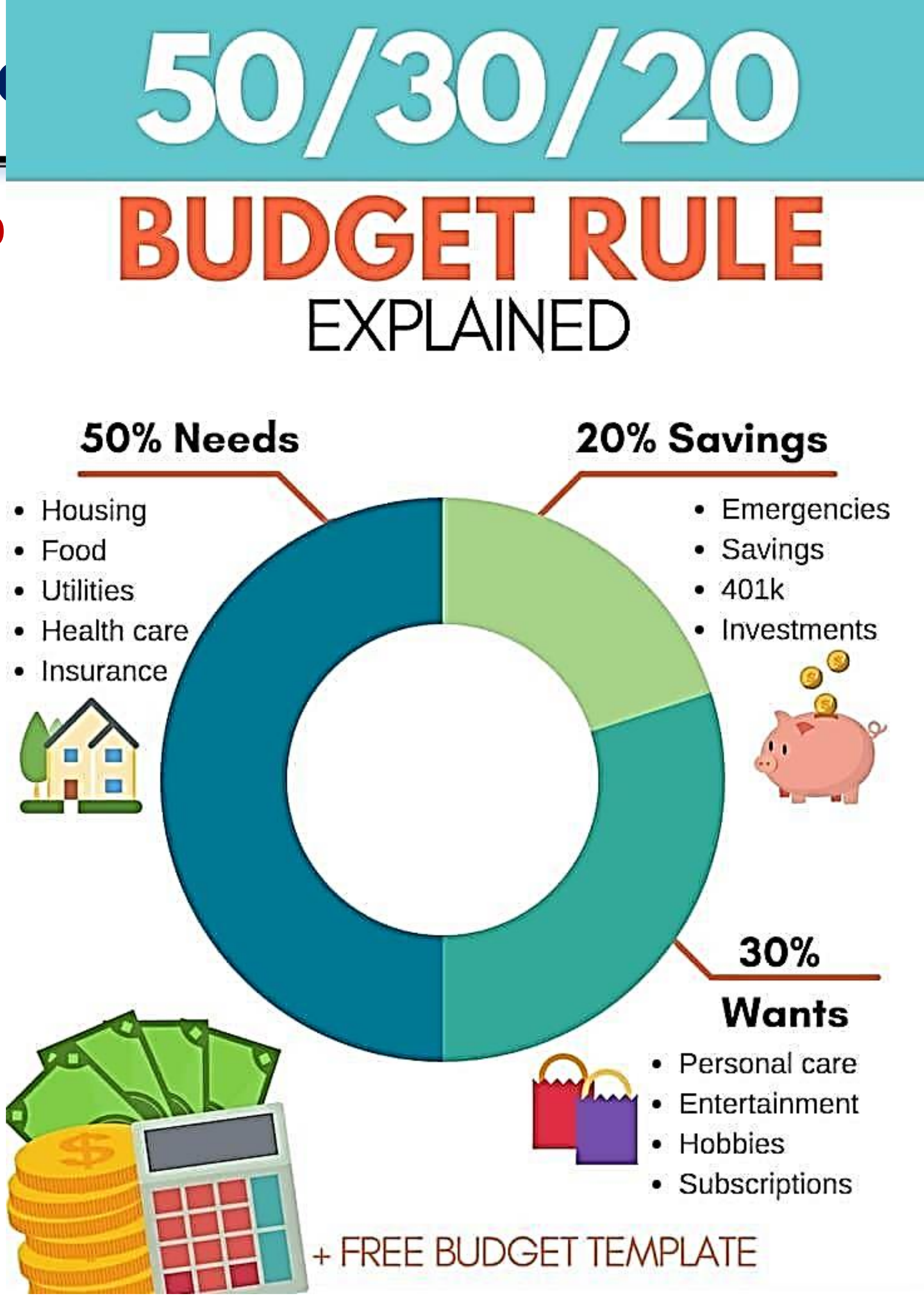


Introduction to Engineering Economics

Introduction to Engineering Eco

50/30/20 Budget Rule





Introduction to Engineering Economics

Introduction to Engineering Economics

Goods

- A grocery store
- Clothing store
- Jewelry store
- Car dealership
- Toy store

Services

- Shipping company
- A mechanic
- Cosmetic surgery
- Barber shop
- Public transit

Combo

- A store that sells and fixes electronics
- A shop that sells and delivers pizza

Introduction to Engineering Economics

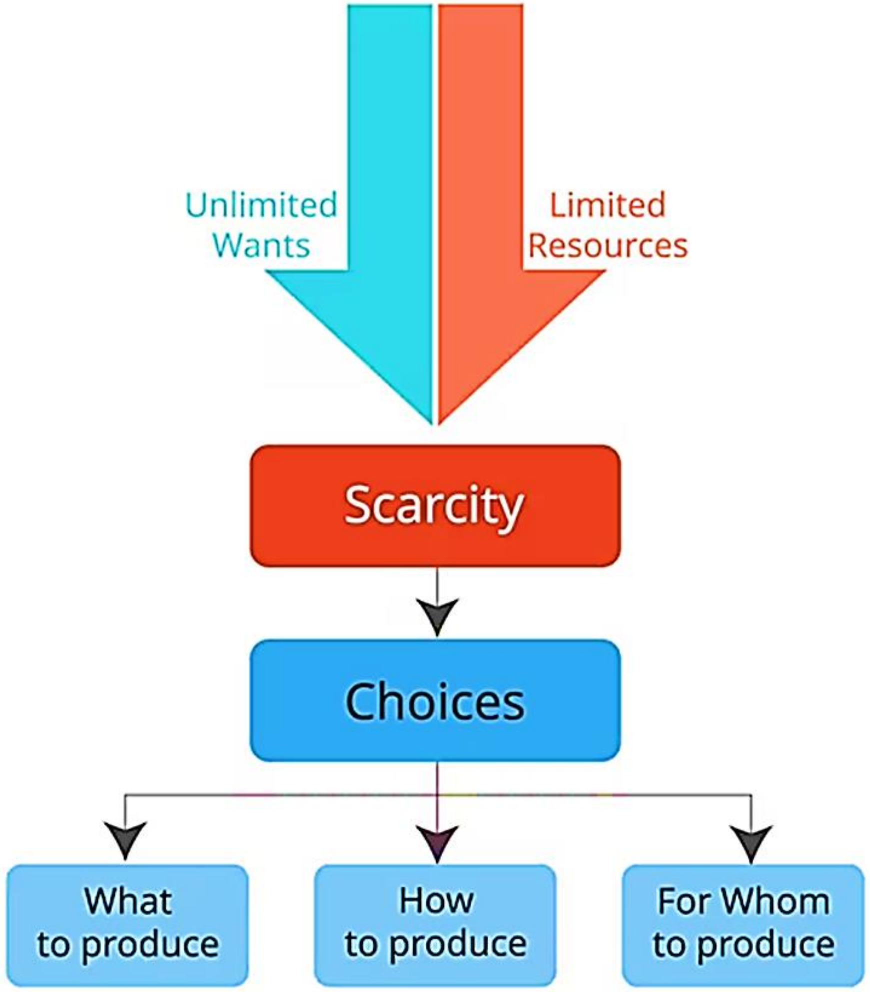
Anything Used to Get or Create Goods and Services is Called Resources

Introduction to Engineering Economics

Factors of Production

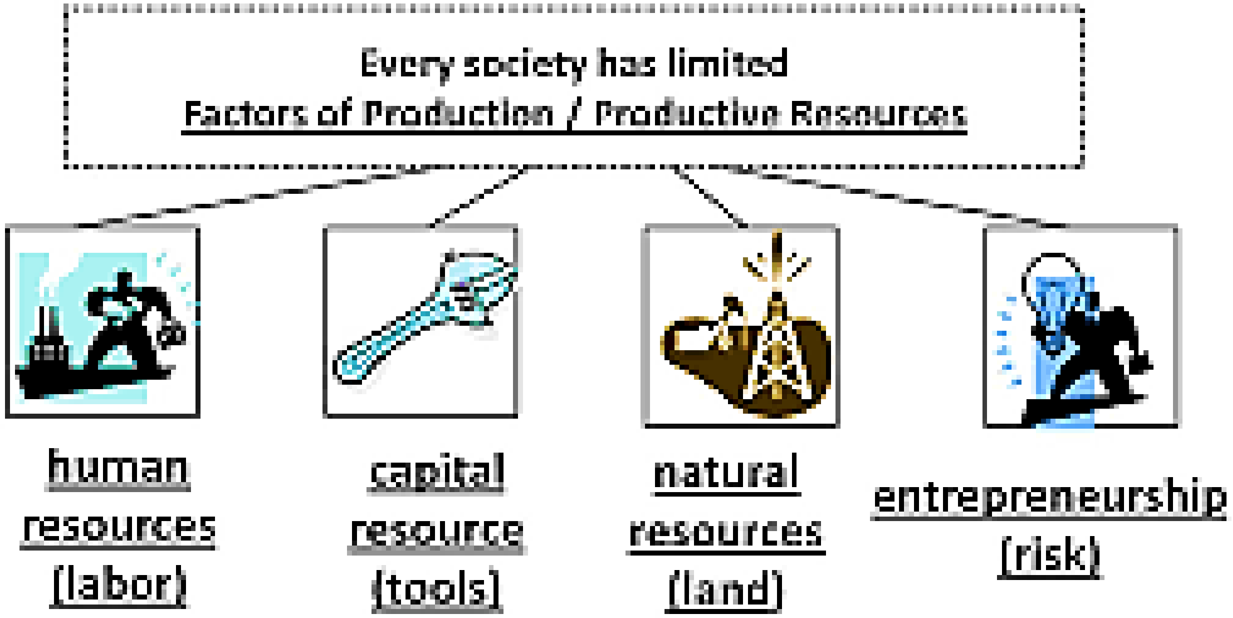
- **The resources used to produce goods and services; also known as production inputs.**
- **Natural resources/Land**
  - Resources provided by nature and used to produce goods and services
- **Labor**
  - The physical and mental effort people use to produce goods and services
- **Human capital**
  - The knowledge and skills acquired by a worker through education and experience.
- **Physical capital & Financial Capital**
  - The stock of equipment, machines, structures, and infrastructure that is used to produce goods and services.
  - Input of funds
- **Entrepreneurship**
  - The effort used to coordinate the factors of production—natural resources, labor, physical capital, and human capital—to produce and sell products.

Introduction to Engineering Economics

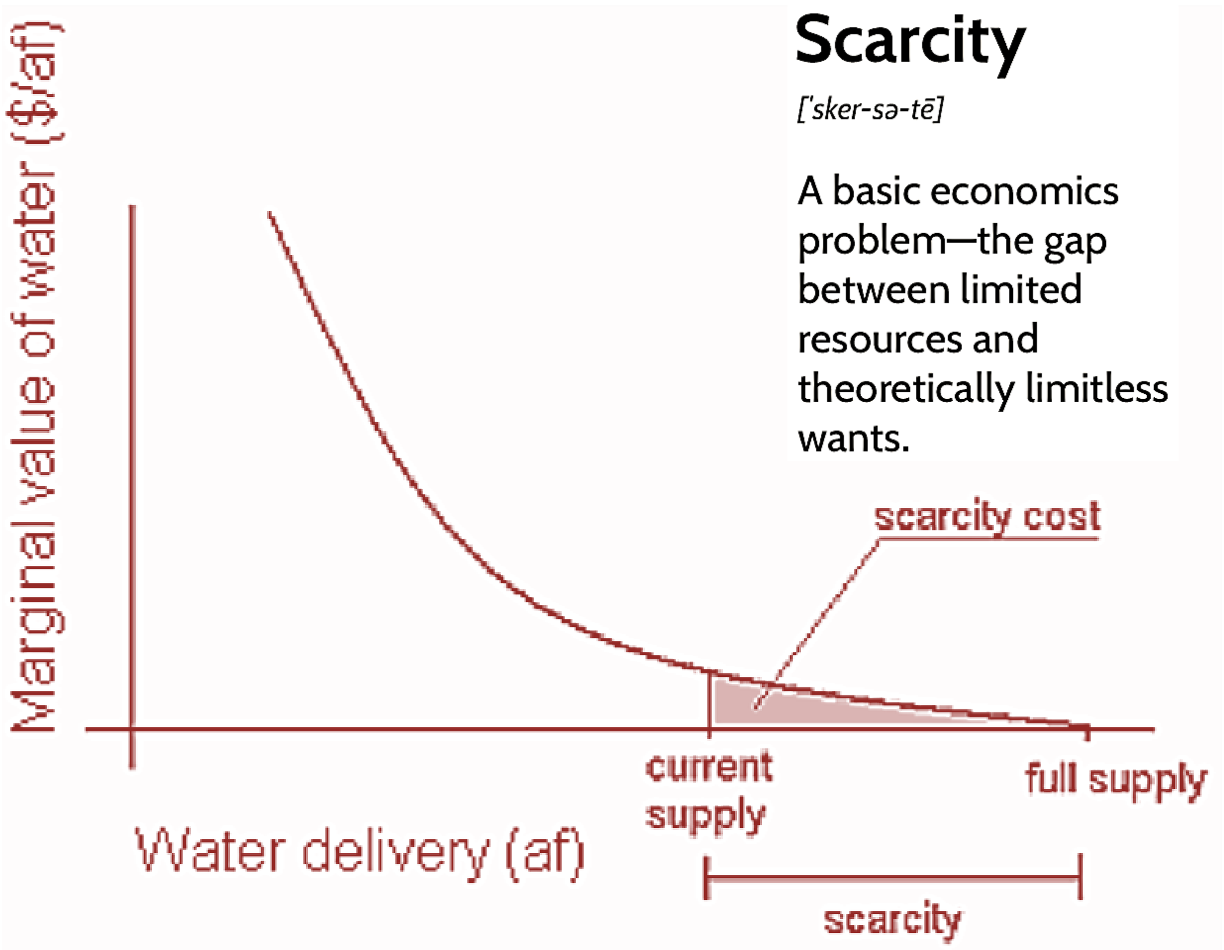


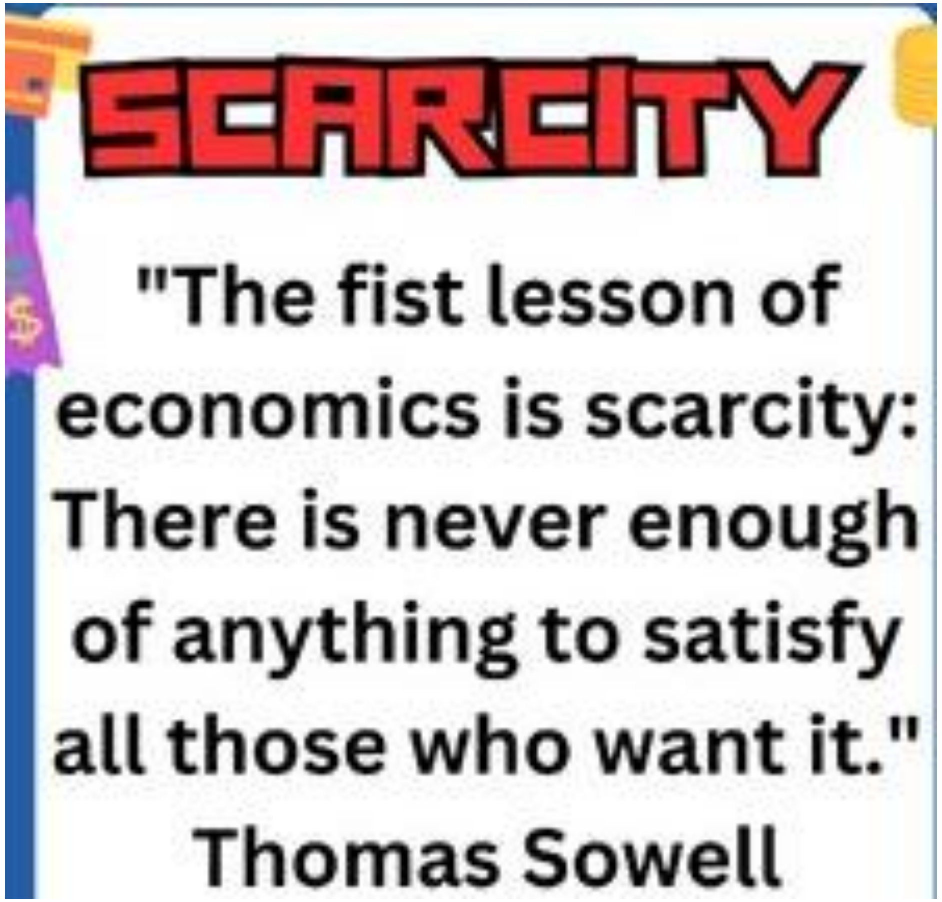
What is Economics?

Economics is about how people choose to use their limited or scarce resources to satisfy their unlimited wants.



Introduction to Engineering Economics





Introduction to Engineering Economics

Limited Resources

SCARCITY of goods and services

unlimited wants

ECONOMIZING PROBLEM
(must make choices)

society's 3 options for dealing with scarcity

Economic Growth

Improve the use of available resources

Reduce Expectations